DeepTMHMM vs TMHMM on fungal proteins

BFD functional annotation benchmark, 2026-11-06.

Summary. DeepTMHMM separates signal peptides from transmembrane (TM) helices much better than TMHMM. It misses many single-pass anchors, mostly type II signal anchors, which it calls signal peptides. A simple rule, TMHMM plus a SignalP 6 filter, keeps most of DeepTMHMM's signal-peptide gain without that loss. TMHMM and SignalP 6 are already computed for every BFD genome. DeepTMHMM would cost about 5,400 GPU-hours (estimated) even with the faster driver, so it stays out of this BFD release.

1. What was compared

Tool	Version and settings
TMHMM	2.0, `--short`
DeepTMHMM	1.0 academic, private fast image, `--topo-batch 1` (same output as stock `predict.py`)
SignalP	6.0, eukaryote, fast mode (BFD pipeline settings)
TMHMM+SignalP	TMHMM helices; drop the first helix when SignalP calls a signal peptide and the helix starts at or before the cleavage site + 5

- **SwissProt set (with truth).** 3,741 reviewed fungal UniProt proteins with experimental evidence: multi-pass membrane 966, single-pass membrane 333, peripheral membrane 488, soluble cytosol/nucleus 1,000 (sampled from 3,258), experimental signal peptide 168, experimentally secreted 847, and 160 with experimental helix or topology features.
- **BFD sample (no truth).** Proteins from the current BFD tables, stratified by the TMHMM and SignalP calls, across 19 phyla (Section 4).
- **Held out.** DeepTMHMM was trained on 3,576 UniProt proteins (DeepTMHMM.3line). All results use only proteins with < 30% identity over ≥ 50% of the query to any training protein (MMseqs2). This removed 583 of 3,741 SwissProt proteins. Results on all proteins differ by 2 points or less.

2. SwissProt results

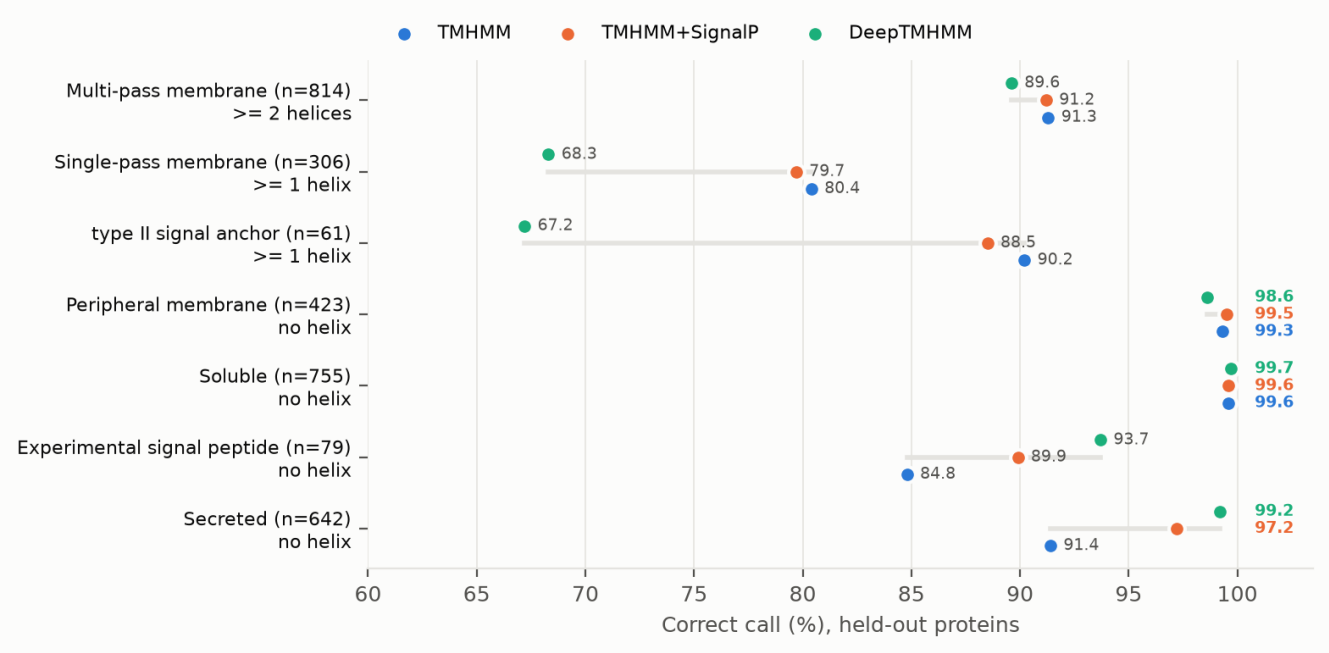


Figure 1: Correct calls per SwissProt set, held-out proteins.

Set (held out)	n	Correct call	TMHMM	TMHMM + SignalP	DeepTMHMM
Multi-pass	814	≥ 2 helices	91.3%	91.2%	89.6%
Single-pass	306	≥ 1 helix	80.4%	79.7%	68.3%
Single-pass, type II	61	≥ 1 helix	90.2%	88.5%	67.2%
Peripheral membrane	423	no helix	99.3%	99.5%	98.6%
Soluble	755	no helix	99.6%	99.6%	99.7%
Experimental signal peptide	79	no helix	84.8%	89.9%	93.7%
Secreted	642	no helix	91.4%	97.2%	99.2%
Experimental helices, all placed	116	exact	50.0%	50.0%	45.7%
Experimental N-terminus side	95 / 84	in or out	63.2%	not scored	79.8%

“All placed” = every experimental helix overlaps a predicted helix by ≥ 5 residues and the counts match. SignalP 6 alone calls a signal peptide in 96.2% of the experimental signal-peptide set and 95.0% of the secreted set. DeepTMHMM calls SP in 100% and 98.4%.

3. Where each tool fails

- **TMHMM calls signal peptides TM helices.** It calls a helix in 8.6% of held-out secreted proteins and 15.2% of the experimental signal-peptide set. The SignalP rule cuts the secreted error from 8.6% to 2.8%.

- **DeepTMHMM calls type II signal anchors signal peptides.** It finds no helix in 32.8% of type II single-pass proteins (TMHMM: 9.8%). SignalP 6 does not make this mistake, so the SignalP rule loses only 1 of 61.
- **DeepTMHMM calls many single anchors globular.** It misses 97 of 306 held-out single-pass proteins; it calls 74 of these GLOB and 23 SP. Type I (n = 47) and type IV tail anchors (n = 25) score the same for both tools (95.7%, 72.0%).
- **DeepTMHMM predicts topology better.** The N-terminus side is right in 79.8% of experimental cases, against 63.2% for TMHMM.
- **DeepTMHMM splits helices.** 22% of BFD proteins with helices (644 of 2,892) have two DeepTMHMM helices separated by a loop of 3 residues or fewer. Merging these pairs does not help on SwissProt: multi-pass (≥ 2 helices) falls from 89.6% to 86.9%, and exact helix counts rise only from 50.0% to 51.7%. Some short loops are real hairpins.
- **Both tools place every helix correctly in only about half** of the 116 held-out proteins with experimental helix positions.

4. BFD sample: agreement between the tools

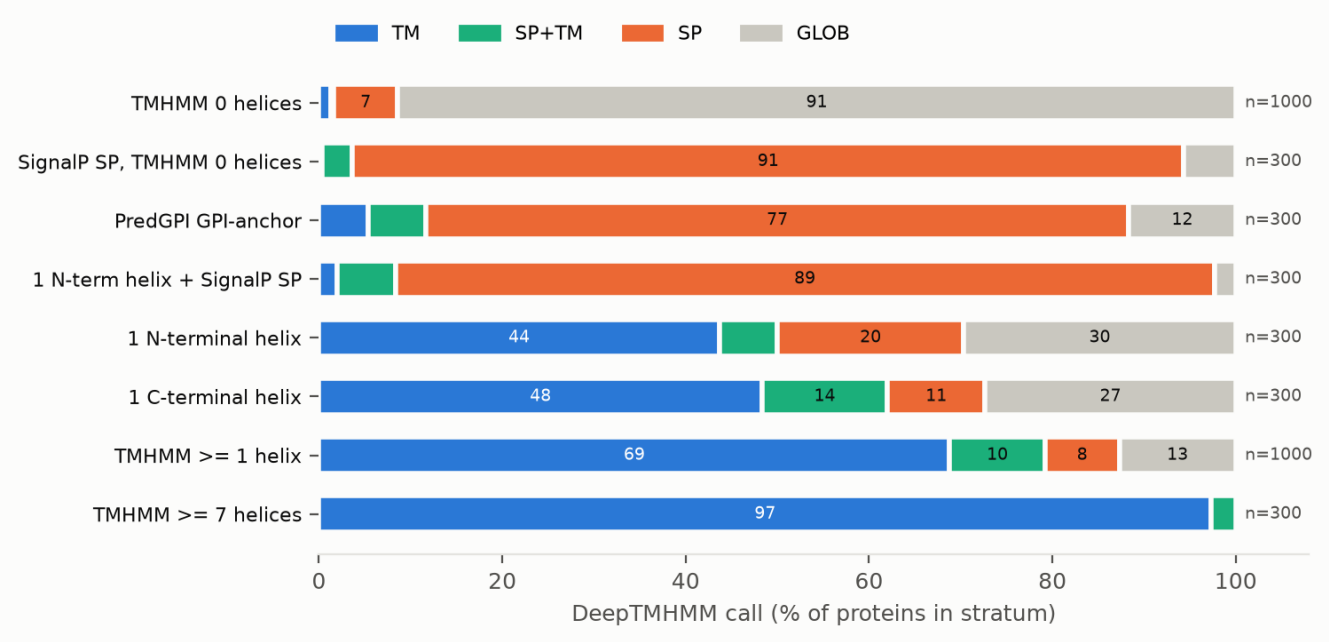


Figure 2: DeepTMHMM calls per BFD stratum (held-out proteins). Numbers are percent; labels below 7% are omitted.

Stratum (held out)	n	Same helix count as TMHMM	SignalP SP
TMHMM 0 helices	1,000	98.4%	5.7%
SignalP SP, TMHMM 0 helices	300	96.3%	100%
PredGPI GPI-anchor	300	76.7%	77.3%
1 N-terminal helix + SignalP SP	300	7.7%	100%
1 N-terminal helix	300	40.0%	12.3%
1 C-terminal helix	300	54.3%	19.7%
TMHMM ≥ 1 helix	1,000	40.0%	12.0%
TMHMM ≥ 7 helices	300	48.7%	2.0%

- The tools agree on proteins without helices.

- In the “1 N-terminal helix + SignalP SP” stratum, DeepTMHMM and SignalP agree that 96% of TMHMM’s helices are signal peptides. This is the case the SignalP rule corrects.
- In the “1 N-terminal helix” stratum without a SignalP peptide, DeepTMHMM calls 20% SP. The SwissProt type II result and the expert review (Section 5) say most of these are signal anchors, and TMHMM is right.
- For proteins with 7 or more helices, both tools agree on multi-pass, but the exact helix counts agree for only 48.7%.

5. Expert review of 80 disagreements

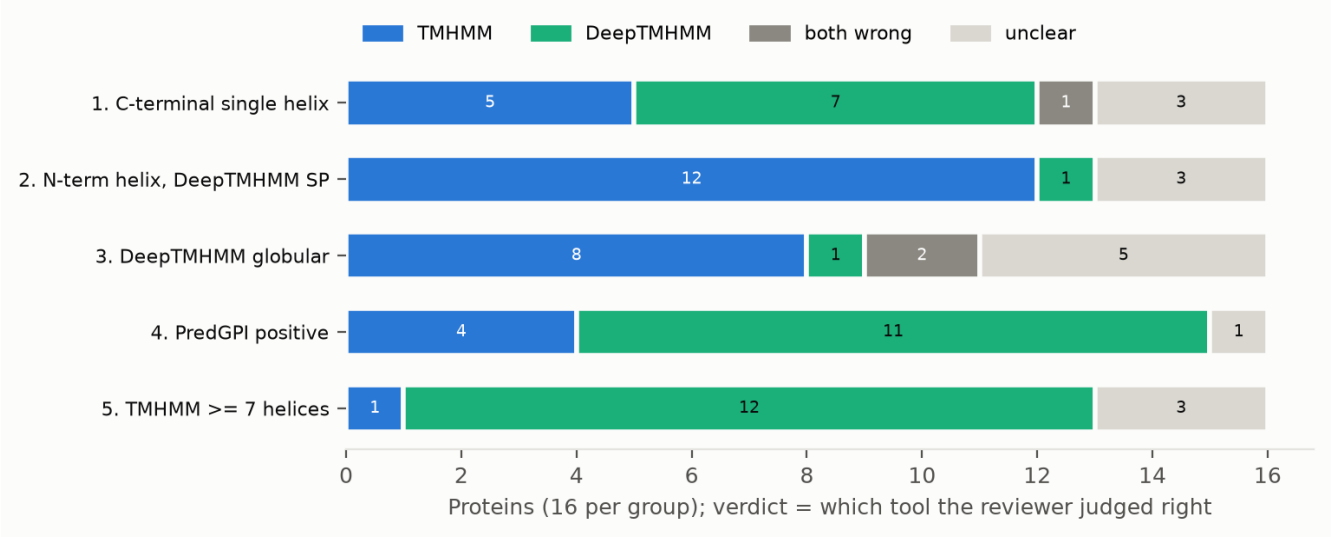


Figure 3: Verdicts of the expert review, 16 BFD proteins per disagreement group.

An agent acting as a fungal membrane-protein expert reviewed 80 BFD proteins (16 per group) where the tools disagree. It used the sequence, Pfam domains (i-Evalue < 1e-3) and the best SwissProt hit. There is no experimental truth for these proteins, and the groups were chosen because the tools disagree, so these counts are not accuracy rates.

Group	TMHMM right	DeepTMHMM right	Both wrong	Unclear
1. C-terminal single helix	5	7	1	3
2. N-terminal helix, DeepTMHMM calls SP	12	1	0	3
3. DeepTMHMM calls globular	8	1	2	5
4. PredGPI positive	4	11	0	1
5. TMHMM ≥ 7 helices	1	12	0	3
Total	30	32	3	15

Patterns the reviewer reported:

- **Group 2 (TMHMM right):** most are type II signal anchors, for example Golgi glycosyltransferases (OCH1, MNN9, GT25, GT49, GT90, Fringe), Grx6/7, MAPR and V-ATPase subunit e.
- **Group 3 (TMHMM mostly right):** real N-terminal anchors (P450, b5 reductase, Tom20) and tail anchors called globular by DeepTMHMM. TMHMM’s typical false positive here is a GxGxxG dinucleotide-binding motif.
- **Group 4 (DeepTMHMM right):** TMHMM calls the signal-peptide core and the C-terminal GPI attachment signal helices. The GPI signal is cleaved, so it is not a TM helix.

- **Group 5 (DeepTMHMM right):** DeepTMHMM finds 12 helices in all 8 MFS transporters; TMHMM finds 8 to 10.
- **Group 1 (mixed):** TMHMM is right for tail anchors (Tom7, metaxin); DeepTMHMM is right for GPI signals (yapsins). Several are spurious short ORFs.
- **The SignalP rule** helped in all 6 cases where it applied and hurt in none. A rule based on DeepTMHMM's SP call alone would remove about 12 real type II anchors in group 2.
- **Gene models:** 5 look like wrong start codons, 3 like fusions, and 6 like spurious short ORFs (details in `review/expert_review.tsv`).

6. Run time

DeepTMHMM runs ESM-1b and then 5 topology models, one protein at a time. The private fast driver (`predict_fast.py`) batches both steps.

DeepTMHMM run, *V. dahliae* 463 (9,789 proteins), RTX 6000 Ada	Wall time	Residue labels identical to stock
stock `predict.py`	57 min	reference
fast, topology batch 1	55-58 min	99.99-100%
fast, topology batch 4	22 min	99.84%
fast, topology batch 8	16 min	99.54%
fast, topology batch 16	12.7 min	99.34%
stock `predict.py`, 8 CPU cores, 300 proteins	3 h 17 min	(about 110x slower than GPU)

- A full BFD pass (about 261 M proteins, estimated from proteome file sizes) would take about 25,000 GPU-hours with the stock driver, or about 5,400 GPU-hours at topology batch 16 (estimated).
- TMHMM took 78-160 s and SignalP 6 took 20-28 s on the same 2,300-4,200 protein BFD benchmark sets.

7. Recommendation and open questions

1. Keep TMHMM in BFD. Add the SignalP rule as a derived column or view: a TMHMM helix that overlaps a SignalP 6 signal peptide is not counted as TM.
2. Do not run DeepTMHMM on all of BFD now. If its topology (N-terminus side) is needed for a subset, use the fast image at topology batch 4 (2.6x faster, 99.84% identical labels).
3. Treat GPI-anchor predictions without a signal peptide with caution.
4. Open: should helix counts for multi-pass proteins come from DeepTMHMM? The expert review favours it for MFS transporters, but SwissProt multi-pass accuracy is slightly lower than TMHMM's (89.6% vs 91.3%).
5. Open: the expert review is one reviewer on 80 proteins with no experimental truth. A second reviewer or a curated topology set would firm up groups 1 and 3.

8. Files and methods

All paths are under `analysis/deeptmhmm_vs_tmhmm/` in `Fungi_BFD_runs`.

File	Content
`swissprot/build_bench.py`, `swissprot/labels.tsv`	UniProt queries and labels; downloads dated 2026-11-06
`sample_bfd.sh`, `sample_bfd_neg.py`, `sample_bfd_gpi.sh`	BFD stratified sampling (seed 20261106)

File	Content
`run_tools.sh`	TMHMM, SignalP 6 and DeepTMHMM on one set
`analyze.py`, `combined_rule.py`	Scoring; `results/summary.txt`, `results/combined_rule.txt`
`results/*_per_protein.tsv`	Per-protein calls
`review/expert_review.tsv`, `review/expert_review.md`	Expert review
`make_figures.py`, `figures/`	Figures (PNG and PDF)

The DeepTMHMM training set, sequence files and tool outputs are kept on HPCC and not committed. `predict_fast.py` and the fast image stay private under the DeepTMHMM academic license.